Rabeya Jamshad

Computer Science and Engineering
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Research Interests Robotics, Human-Robot Interaction, Behavior Modeling, Reinforcement Learning Machine Learning, Healthcare Technology, Action Teams, Team Dynamics

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Education

University of California San Diego 2021 - 2027 (Expected)
Ph.D., Computer Science and Engineering
Advisor: Dr. Laurel Riek

Georgia Institute of Technology 2017 - 2019
M.Sc., Electrical and Computer Engineering
Advisor: Dr. Ayanna Howard

Lahore University of Management Sciences 2011 - 2015
B.S., Electrical Engineering
Advisor: Dr. Abubakr Muhammad

Selected Publications

Jamshad, R., Ramaraj, P., Masanam, D., & Riek, L. D. (2026). Between Autonomy and Occupational Identity: How Emergency Responders Reimagine Robot Roles in Firefighting & Rescue Work. In *Prep.*

Jamshad, R., Pal, M., Gupta, V., and Riek, L. D. (2026). *Title anonymized for review.* In IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026). In *Review.*

Jamshad, R., Haripriyan, A., Sonti, A., Simkins, S. and Riek, L.D., (2026). Human-robot action teams: How robot proactivity affects teamwork. *Computers in Human Behavior: Artificial Humans.*

Zhang, T., Garrison, C., Hockenberry, K., **Jamshad, R.,** Riek, L. D., & Simkins, S. (2026). Human-Robot Teaming in Nursing Simulations: Team Member Communication and Performance Effects. In *Companion Proc. of the ACM/IEEE International Conference on Human-Robot Interaction (HRI).*

Jamshad, R., Simkins, S., and Riek, L. D. (2025). Proactive Robots in Human Robot Action Teams. *AI in Teams.* In *Press*.

Haripriyan, A., **Jamshad, R.,** Ramaraj, P., and Riek, L.D. (2024) Human-Robot Action Teams: A Behavioral Analysis of Team Dynamics. In *Proceedings of the 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).*

Jamshad, R., Haripriyan, A., Sonti, A., Simkins, S., and Riek, L.D. (2024) "Human-Robot Action Teams: How Robots Can Be Proactive Teammates". In *Companion Proc. of the ACM/IEEE International Conference on Human-Robot Interaction (HRI).*

Matsumoto, S., **Jamshad, R.,** Ghosh, P., and Riek, L. D. (2023). Robot, Uninterrupted: Telemedical Robots to Mitigate Care Disruption. In *Proceedings of the ACM/IEEE International Conference on Human Robot Interaction (HRI).* [Acceptance rate: 25%]

Jamshad, R., Fry, K.E., Chen, Y.P. and Howard, A., (2019). Design of a robotic crib mobile to support studies in the early detection of cerebral palsy: A pilot study. In *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).*

Experience

Ph.D. Student Researcher, UC San Diego 2021 - Present
Designed and developed proactive robot systems to support human-robot teaming

- Developed and validated intent inference method for multi-party human-robot teams
- Developed and investigated proactive robot interaction behaviors to support fluent collaboration in human-robot teams
- Led and conducted multiple large-scale studies with humans and physical robots to validate new robot designs and behaviors
- Collected and analyzed multimodal human-robot interaction and behavior data

Senior Robotics Engineer, Xavor Corporation 2020 - 2021
Developed navigation control system of a mobile service robot for older adults

- Developed and integrated a multimodal autonomous navigation system in ROS of a companion robot for older adults
- Integrated machine learning models with the robot's control system

M.Sc. Student Researcher, Georgia Institute of Technology 2018 - 2019
Developed a robot for infants with neuromuscular disorders

- Developed a robotic crib mobile to facilitate the early diagnosis of neuromuscular disorders in infants
- Validated the effectiveness of the robotic intervention through at-home experiments

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Professional Competencies

Software and Programming Languages
C++, Python, ROS, Webots, Matlab, Linux, Arduino, Latex, NumPy, PyTorch

Control Design and Robotics
Control Systems | Mobile Robotics | Search and Optimization | Artificial Intelligence | Sensor Fusion | Human-Robot Interaction | Reinforcement Learning

Research Methods
Designing, conducting, and analyzing controlled experiments with human subjects; Qualitative and quantitative data collection and analysis; Rapid prototyping

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Leadership

Voting Graduate Student Member on the Chancellor's Advisory Committee on the Status of Women 2022 - 2024

Secretary for RoboGrads at UC San Diego 2022 - 2024

Panelist on USEFP STEM Careers for Women 2020

General Secretary for the Pakistan Student Association (PSA) 2018 - 2019

Academic Service

Reviewer: THRI, InGroup, HRI, SGR, ICSR

Student Mentoring

Graduate and Undergraduate Mentor, UC San Diego 2023 - 2026

- Advika Sonti (B.S. Cognitive Science)
- Priyanshu Arora (M.Sc. Computer Science)
- Arthi Haripriyan (M.Sc. Computer Science)
- Deep Masanam (Ph.D. Computer Science)

Invited Talks and Demos

InGroup 2024 Symposium, HRI 2024 Poster, InGroup 2023 Poster, UC San Diego Open House 2022-2023, Academy of Management 2025

Awards and Honors

Powell Fellowship 2021
Fulbright Scholar 2017